

Day 21 – AWS Serverless Web Application

Introduction

Serverless architecture removes the need for managing servers manually.

AWS Lambda

Runs backend code automatically using events.

API Gateway

Handles secure API communication.

DynamoDB

Stores NoSQL application data.

Amazon S3

Hosts frontend static websites.

CloudWatch

Monitors serverless applications.

Real-World Project

Student portal using Lambda, API Gateway, and DynamoDB.

Security

IAM Roles, HTTPS APIs, encryption, API throttling.

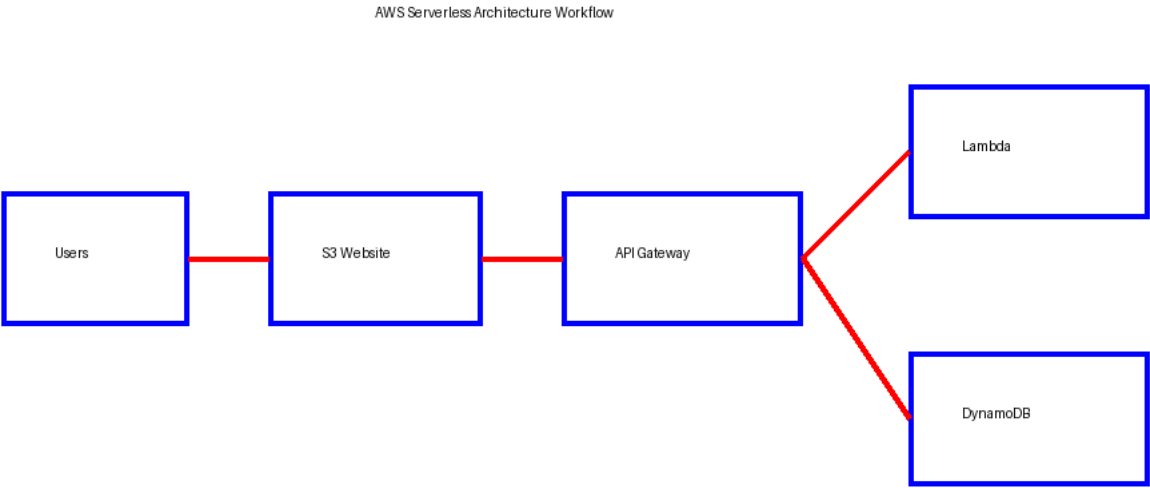
Advantages

Automatic scaling, lower cost, high availability.

Use Cases

SaaS products, AI APIs, mobile apps, IoT systems.

Serverless Workflow Diagram



AWS Serverless Components

Service	Purpose
Lambda	Backend processing
API Gateway	API communication
DynamoDB	NoSQL storage
S3	Frontend hosting
CloudWatch	Monitoring & logging

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